

Digital Memory Search Engine

A Digital Memory Search Engine is an AI system that stores person's digital information and allow that to search it later using nature language.

For example, imagine you have:

- Class notes
- PDFs
- YouTube videos
- Web articles
- Emails
- Personal notes
- Chat conversations
- Images
- Documents

Normally, these are scattered across many apps and folders.

My project brings all of them into one place and lets users ask questions in simple English.

- **Example Scenario**

Suppose during your semester you:

- Watched a YouTube video about Dynamic Programming.
- Saved a PDF on Computer Networks.
- Wrote notes about MongoDB.
- Had a discussion with friends about a project.

After 6 months, you forget where you saved something.

Instead of manually searching everywhere, you ask:

"Show me all notes related to Dynamic Programming."

AI instantly finds:

- ✓ PDF pages

✓ YouTube videos

✓ Personal notes

✓ Chat messages

- **Another Example**

You previously watched a YouTube video explaining Binary Search.

Months later, you ask:

"Which video explained Binary Search with animations?"

The system returns:

- Video title
- Thumbnail
- Timestamp
- Summary
- Important concepts

Real-Life Use Cases

Student

A student asks:

"Find everything I studied about Operating Systems."

System shows:

- Notes
- PDFs
- Videos
- Previous questions
- Important formulas

Developer

Developer asks:

"Show me the React authentication code I saved last month."

System finds:

- Code snippets
- GitHub links
- Notes

Researcher

Researcher asks:

"Find all papers related to Large Language Models."

System shows all saved papers and summaries.

How It Works Internally

Step 1: Collect Data

User uploads:

- PDFs
- DOCX files
- Images
- Notes
- YouTube URLs
- Web links

Step 2: Extract Content

The system extracts text.

Example:

PDF → Text

Image → OCR Text

YouTube → Transcript

Step 3: AI Understands the Content

AI converts text into mathematical representations called **embeddings**.

Similar topics are stored close together.

For example:

Text	Meaning
"DP"	Dynamic Programming
"Memoization"	Dynamic Programming
"Tabulation"	Dynamic Programming

AI understands these are related.

Step 4: Search

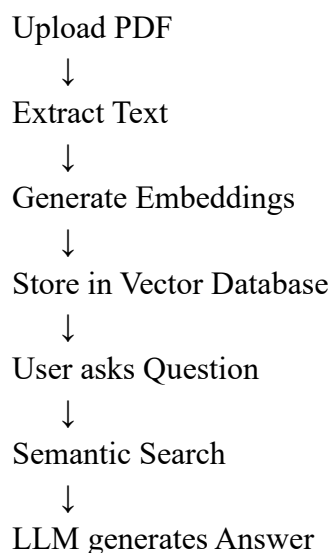
User asks:

"What did I learn about memoization?"

AI searches semantically (by meaning, not exact words).

Even if your notes never contain the exact word "memoization", the system can still find related content.

Example Flow



Example Queries

"What are my notes on DBMS normalization?"

"Find videos related to recursion."

"Show documents about React Hooks."

"What did I save about cybersecurity?"

"Summarize everything about Machine Learning."

"Which YouTube video explained BFS?"

Features for Final Year Project

Basic Features

- User Login/Register
- Upload PDFs and documents
- Add YouTube links
- Save notes
- Search using natural language
- AI summaries

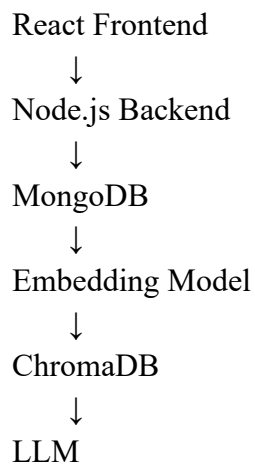
Advanced Features

- Semantic Search
 - Chat with your knowledge base
 - Auto-generated tags
 - Smart recommendations
 - Voice search
 - Topic timeline
 - Spaced repetition reminders
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Suggested Tech Stack

Component	Technology
Frontend	React.js
Backend	Node.js + Express
Database	MongoDB
Vector Database	ChromaDB or Pinecone
AI Model	OpenAI Embeddings / Sentence Transformers
LLM	OpenAI GPT or local LLM
OCR	Tesseract OCR
YouTube Transcript	YouTube Transcript API
Authentication	JWT

Example Architecture



This is a strong final-year project because it combines **Full Stack + AI + NLP + Information Retrieval**, and solves a real problem that students, developers, and researchers face every day.

